DATA PRODUCT CARD

PJATK • Big Data Course • Two-Page Template

Use this template to present your data product in a concise, market-ready format. Fill it manually or with AI support.

1. PRODUCT OVERVIEW

Product name: NYC Yellow Taxi Analytics

Owner / Team: Magdalena Kochanowska

Business domain: Mobility / Transportation Analytics

Primary users: Data analysts, BI analysts, students, researchers analyzing taxi activity and transportation patterns.

Version / update date: v1.0 / 2026-05-10

AT A GLANCE

Format: Parquet

Refresh: Manual / After each pipeline run

Access: Git repository + MS Teams Data Marketplace

Quality metrics: Invalid time records, Negative values, Missing location lookup

Status: Production-like academic project

2. PURPOSE & VALUE

NYC Yellow Taxi Analytics is a processed data product built on NYC Yellow Taxi trip records.
The product enables analysis of taxi activity, revenue, trip volume, and transportation patterns between NYC taxi zones.

The ETL pipeline automatically ingests raw files, validates records, cleans the data, and produces business-ready aggregated datasets in the Gold layer.
The product supports analytical use cases such as revenue analysis, passenger activity analysis, and taxi zone utilization analysis.

The data product reduces the effort required to manually clean and process raw taxi trip data and provides consumers with structured Parquet datasets ready for analytics or visualization tools.

3. DATA SOURCES

List the main source systems or files used to build this product.

Source	Type	Refresh
NYC TLC Yellow Taxi Trip Records	File / Parquet	Manual
NYC Taxi Zone Lookup Table	CSV File	Manual
File Queue Ingestion System	Local Filesystem Queue	Automatic

4. HOW TO ACCESS THIS PRODUCT

- Where is the product published? MS Teams Data Marketplace + Git repository
- Direct link: https://github.com/MagKaur/nyc-taxi-pipeline
- How to download or query it: [Short instruction]
  1. Clone the repository
  2. Run the Dagster pipeline
  3. Read Parquet files from the Gold layer
- Credentials or access notes: [If needed] None
- Example file/table name: data/warehouse/gold/taxi_metrics/

5. QUALITY SNAPSHOT

Metric	Value
Invalid time records	5
Negative values	116144
Missing location lookup	0
Invalid records in silver	116148

6. HIGH-LEVEL FLOW

Summarize the pipeline in one line:

Source	Bronze	Silver	Gold	Product

NYC TLC Dataset
→ data/queue/
→ raw ingestion
→ bronze/yellow_taxi_raw
→ silver/yellow_taxi_clean
→ gold/taxi_metrics
→ Data Marketplace

Detailed Contract & Operating Information

7. SCHEMA OVERVIEW

List the main fields that a consumer needs to understand.

Field name	Type	Description	Example
gold_record_id	string	Stable hash identifier for Gold record	a82f...
month	timestamp	Month of trip activity	2026-01-01
pickup_zone	string	Pickup taxi zone	Midtown Center
dropoff_zone	string	Dropoff taxi zone	JFK Airport
trip_count	long	Number of trips	1520
total_revenue	double	Total trip revenue	45231.25
avg_distance	double	Average trip distance	3.8

8. REFRESH & OPERATIONS

Refresh frequency: After each pipeline run

Update trigger: Dagster Sensor + ETL Pipeline

Storage location: Local filesystem / Parquet files

Expected batch size / volume: Millions of taxi trip records per dataset

Dependencies: NYC TLC datasets, Taxi zone lookup CSV, Dagster, PySpark

9. EXAMPLE USAGE

Business question:
Which NYC taxi zones generate the highest revenue and trip volume? Example query / code / steps:

```
from pyspark.sql import SparkSession

spark =
SparkSession.builder.appName("analytics").getOrCreate()
df = spark.read.parquet(
    "data/warehouse/gold/taxi_metrics"
)
df.orderBy("total_revenue", ascending=False).show()
```

10. LIMITATIONS & RISKS

- The pipeline currently operates locally.
- No real-time streaming support.
- Product quality depends on source NYC TLC data quality.
- Some records contain negative values or invalid timestamps.
- Data is not exposed through API or database access.

11. GOVERNANCE & CONTACT

Owner: Magdalena Kochanowska

Contact: MS Teams / PJATK

Steward / responsible person: Magdalena Kochanowska

Access policy: Public for class / educational usage

Marketplace location: MS Teams → Data Marketplace folder

Feedback